



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

CompTIA Certified Server+ Training

TGS Ref No: TGS-2024048318

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 4.1

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
3.0	1 May 2025	Previous release — CompTIA Certified Server+ (SK0-005) master trainer slides and lesson plan.	Tertiary Infotech Academy
4.0	19 August 2026	Rebuilt courseware from the single-source pipeline: 3-day lesson plan aligned to the 33 hands-on KillerCoda labs across the four SK0-005 exam domains.	Dr Alfred Ang
4.1	19 August 2026	Added Linux/Kubernetes/Docker lab environments and per-lab asset folders (guidelines/cheatsheet/worksheet PDFs, CSV+Excel data, scripts and container/Kubernetes YAML).	Dr Alfred Ang

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Course Information

Course Title	CompTIA Certified Server+ Training
WSQ Course Reference	TGS-2024048318
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	3 days · 8 training hours per day (24 hours)
Daily Timing	9:00 am – 6:00 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on server administration labs on the free Killercoda Ubuntu playground
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Install and manage server hardware — racking, cabling, power and cooling; deploy and manage storage (RAID, shared storage, capacity planning); and maintain hardware with out-of-band management and firmware upgrades.
- LO2: Administer server operating systems and services — OS installation, partitions and file systems, IP/DNS/DHCP/VLANs, server roles, monitoring, high availability, virtualization, scripting and asset management.
- LO3: Secure servers and plan for disaster recovery — data and physical security, identity and access management, mitigation strategies, server hardening, decommissioning, backup and recovery.
- LO4: Troubleshoot server hardware, software, storage, OS and network problems using a structured methodology, and validate disaster-recovery and failover readiness.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on server administration tasks on the Killercoda Ubuntu playground, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 3 from 4:00 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Server Hardware, Storage & OS Administration

Time	Duration	Topic / Activity
9:00–9:30	30 min	Welcome, course introduction, ground rules and mandatory digital attendance (AM)
9:30–10:30	60 min	Topic 1 — Server Hardware Installation and

		Management: form factors, racking, power and cooling; RAID levels and shared storage; out-of-band management and firmware (concepts + demo)
10:30–10:45	15 min	Tea break
10:45–13:00	135 min	Hands-on: Lab 1: Server Form Factors, Racking, and Power Planning; Lab 2: Storage Deployment: RAID 0/1/5/6/10 with mdadm; Lab 3: LVM Provisioning and Capacity Planning; Lab 4: Shared Storage: NAS (NFS/CIFS) and SAN (iSCSI); Lab 5: Out-of-Band Management, BIOS/UEFI, and Firmware
13:00–14:00	60 min	Lunch break
14:00–15:30	90 min	Topic 2 — Server Administration: OS installation, partitions and file systems; IP/DNS/DHCP configuration (concepts). Hands-on: Lab 6: Installing a Server OS (Unattended / Scripted); Lab 7: Partition and Volume Types, File Systems; Lab 8: IP, VLAN, DNS, DHCP, FQDN, and Hosts File
15:30–15:45	15 min	Tea break
15:45–17:45	120 min	Hands-on: Lab 9: Server Firewall and Port Management; Lab 10: Server Roles: Web, File, and Database; Lab 11: Directory Services with OpenLDAP; Lab 12: Performance Monitoring, Baselineing, Event Logs; Lab 13: Data Migration and Transfer
17:45–18:00	15 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 2 — Networking, Virtualization, Security & Disaster Recovery

Time	Duration	Topic / Activity
9:00–9:15	15 min	Day 1 recap and mandatory digital attendance (AM)
9:15–10:30	75 min	Topic 2 (cont.) — High availability, virtualization, scripting and asset management (concepts + demo)
10:30–10:45	15 min	Tea break
10:45–13:00	135 min	Hands-on: Lab 14: High Availability: Clustering, Load

		Balancing, NIC Teaming; Lab 15: Virtualization with KVM / QEMU; Lab 16: Scripting Basics for Server Administration; Lab 17: Asset Management, Documentation, and Licensing
13:00–14:00	60 min	Lunch break
14:00–15:30	90 min	Topic 3 — Security and Disaster Recovery: data and physical security, IAM, mitigation and hardening (concepts). Hands-on: Lab 18: Data at Rest (LUKS) and Data in Transit (TLS/SSH); Lab 19: Physical and Environmental Security Walk-Through; Lab 20: Identity & Access Management for Server Administration; Lab 21: Multi-Factor Authentication with TOTP (PAM)
15:30–15:45	15 min	Tea break
15:45–17:45	120 min	Hands-on: Lab 22: Auditing, Logging, and SIEM Basics; Lab 23: OS and Application Hardening; Lab 24: Patch and Update Management; Lab 25: Secure Decommissioning and Media Destruction; Lab 26: Backup Strategy: Full, Incremental, Differential, Snapshot; Lab 27: Disaster Recovery: Replication and Site Failover
17:45–18:00	15 min	Day 2 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 3 — Troubleshooting & Final Assessment

Time	Duration	Topic / Activity
9:00–9:15	15 min	Day 2 recap and mandatory digital attendance (AM)
9:15–10:30	75 min	Topic 4 — Troubleshooting: the CompTIA 6-step methodology; hardware, storage, OS and network faults (concepts + demo)
10:30–10:45	15 min	Tea break
10:45–13:00	135 min	Hands-on: Lab 28: Troubleshooting Methodology Walk-Through; Lab 29: Troubleshooting Common Hardware Failures; Lab 30: Storage Troubleshooting; Lab 31: OS and Software Troubleshooting; Lab 32:

		Network Connectivity Troubleshooting; Lab 33: Security Troubleshooting
13:00–14:00	60 min	Lunch break
14:00–15:30	90 min	Course revision, exam preparation and CompTIA Server+ practice exam walkthrough
15:30–15:45	15 min	Tea break
15:45–16:00	15 min	Briefing for Assessment
16:00–17:00	60 min	Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book
17:00–18:00	60 min	Practical Performance (PP) — hands-on server administration tasks on the Killercode Ubuntu playground, 1 hour, open book. PM digital attendance

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to exam skill areas)

Topic / Exam skill area	Weighting	Labs
Topic 01: Server Hardware Installation and Management	18%	Lab 1, Lab 2, Lab 3, Lab 4, Lab 5
Topic 02: Server Administration	30%	Lab 6, Lab 7, Lab 8, Lab 9, Lab 10, Lab 11, Lab 12, Lab 13, Lab 14, Lab 15, Lab 16, Lab 17
Topic 03: Security and Disaster Recovery	24%	Lab 18, Lab 19, Lab 20, Lab 21, Lab 22, Lab 23, Lab 24, Lab 25, Lab 26, Lab 27
Topic 04: Troubleshooting	28%	Lab 28, Lab 29, Lab 30, Lab 31, Lab 32, Lab 33